USRP Studies

Week 3: A Little Automata Theory and Spectral Graph Theory

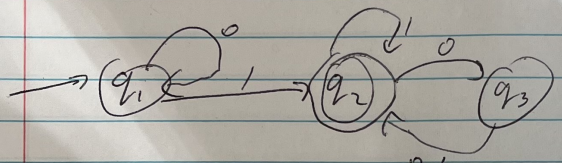
Overview

I spent time researching about specific subfields in TCS and various topics. I also started studying these specific topics more rigorously. I picked up Michael Sipser's "Introduction to the Theory of Computation" and continued watching the TCS toolkit lectures. I skipped ahead for now because the topic was interesting. My goal now, after learning more, is to rigorously study classical models of computation and complexity to better understand quantum computation, while also looking into interesting fields like Spectral Graph Theory. I don't really mind the applications yet.

Automata Theory (Sipser's Intro to ToC)

(1) defn. A finite automaton is a 5-tuple $(Q, \Sigma, \delta, q_0, F)$, where

1. Q is a finite set called the states
2. Σ is a finite set called the alphabet
3. $\delta : Q \times \Sigma \to Q$ is the transition function
4. $q_0 \in Q$ is the start state
5. $F \subseteq Q$ is the set of accept states



$Q = \{q_1, q_2, q_3\}$, $\Sigma = \{0, 1\}$, δ:

	0	1
q_1	q_1	q_2
q_2	q_3	q_2
q_3	q_2	q_2

q_1 start state

$\{q_2\}$ accept states

(1.1) rmk Let A be the set of all strings that machine M accepts. A is the language of M, $L(M) = A$.

Spectral Graph Theory (Lec 13)

Let $G=(V,E)$ be an undirected graph. Sps it is:
- finite
- parallel edges, self loops ok
- no isolated vertices ($\deg v = 0$)

We will study:
- Labellings of V by real #'s ($f: V \to \mathbb{R}$)
  - ↳ We can think of f as a vector:
    $f: V \to \mathbb{R} \equiv (f(v_1), f(v_2), \ldots, f(v_n))$. (isomorphism)

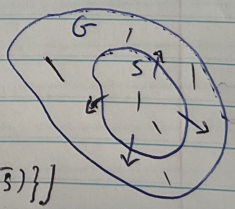
(0.1) rmk $f+g$ ok, cf ok. $\therefore \{f: V \to \mathbb{R}\}$ is a vec. space, with dim. $|V|$.

(1) defn (Local variance) $\mathcal{E}[f] := \frac{1}{2} \cdot \mathop{\mathbb{E}}_{u \sim v}[(f(u)-f(v))^2]$

(1.1) rmk This is a very important qty in Spectral Graph Theory, and we will spend a fair amt of time studying it.

(2) prop (Facts about the local variance)
- $\mathcal{E}[f] \geq 0$
- $\mathcal{E}[cf] = c^2 \mathcal{E}[f]$
- $\mathcal{E}[f+c] = \mathcal{E}[f]$

Intuition:
$\mathcal{E}[f]$ is small $\Leftrightarrow$ f's values don't vary much along edges



Example Let $S \subseteq V$, $I_S(u) = \begin{cases} 1 & \text{if } u \in S \\ 0 & \text{if } u \notin S \end{cases}$.

Then, $\mathcal{E}[f] = \frac{1}{2} \mathop{\mathbb{E}}_{u \sim v}[(I_S(u)-I_S(v))^2]$
$= \frac{1}{2} \cdot \mathop{\mathbb{E}}_{u \sim v}[I\{(u,v) \text{ "crosses" the "cut" } (S, \bar{S})\}]$
$= \frac{1}{2} \{\text{frac. of edges on } \partial S\}$
$= \Pr_{u \sim v}[u \to v \text{ is stepping out of } S]$.